

When transporters own the road, how much can the network carry in a day?

The road-occupancy model and the network bound: thesis chapter 2 and the bound section of chapter 3

A capacity bound for shipyard road networks with shareable widths and asymmetric loaded and empty occupancy

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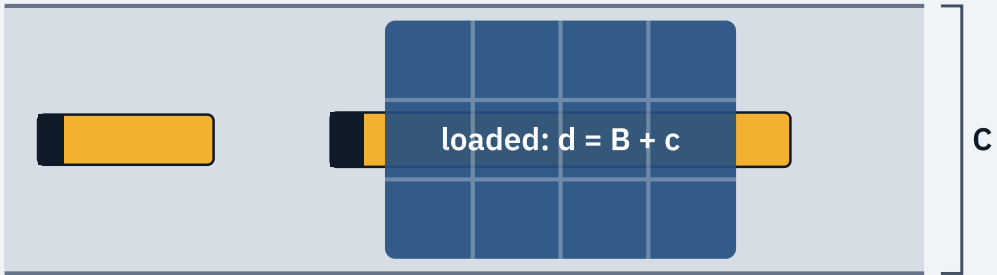
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The system bound and the route-free bound; none of over 50,000 occupancy simulation points in two yards exceeds the bound

One resource = one width container

At every moment: widths in use $\leq$ capacity

capacity $C = \text{road width } W + \text{lateral clearance } c$



empty: not enough width left, waits

This resource over one day ($H = 16 \text{ h}$)



$$\sum \ell \cdot u(d / C) \leq H$$

One row per resource: this is the LP of the network bound

This sub-study in one minute

1

The question

When loaded transporters hold segments and junctions exclusively, how many tasks a day can a yard's network complete? The bound should hold for a whole class of traffic rules and show where it binds.

2

The model

Segments, junctions and stops are width containers, held for a blocking time (passage plus clearance, plus the turn sweep at junctions); loaded and empty vehicles take different widths.

3

Network bound

The occupancy-rate LP plus width bin-packing rows, crane rows and a window edge term. System bound (no sampling noise): Okpo 1,736, Yantai 1,298 a day (main scenario); none of over 50,000 simulation points exceeds it.

4

Route-free bound

Loaded routes are chosen by the LP too, so the bound holds for any routing rule, with a certificate that does not rest on solver tolerances. Route slack in the bound runs from 0 (an entrance binds) to +45% (a through corridor).

The bound tops the capacity interval and names where it binds; a less conservative deadlock-free rule re-checks the lower edge.

The order of this briefing (click to jump)

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The question: write road occupancy as a structural capacity constraint and compute the daily throughput the network can carry

Place in the thesis

Chapter 2: the occupancy model. Chapter 3, bound section: the network bound and its checks, system and route-free bounds, route slack.

Two ceilings and the capacity interval

Throughput $\leq \min\{\text{fleet bound } K \cdot H / \bar{c}, \text{network bound}\}$. The network bound follows from the blocking-time structure and is computed by LP. Attainable capacity lies in [best deadlock-free rule, network bound].

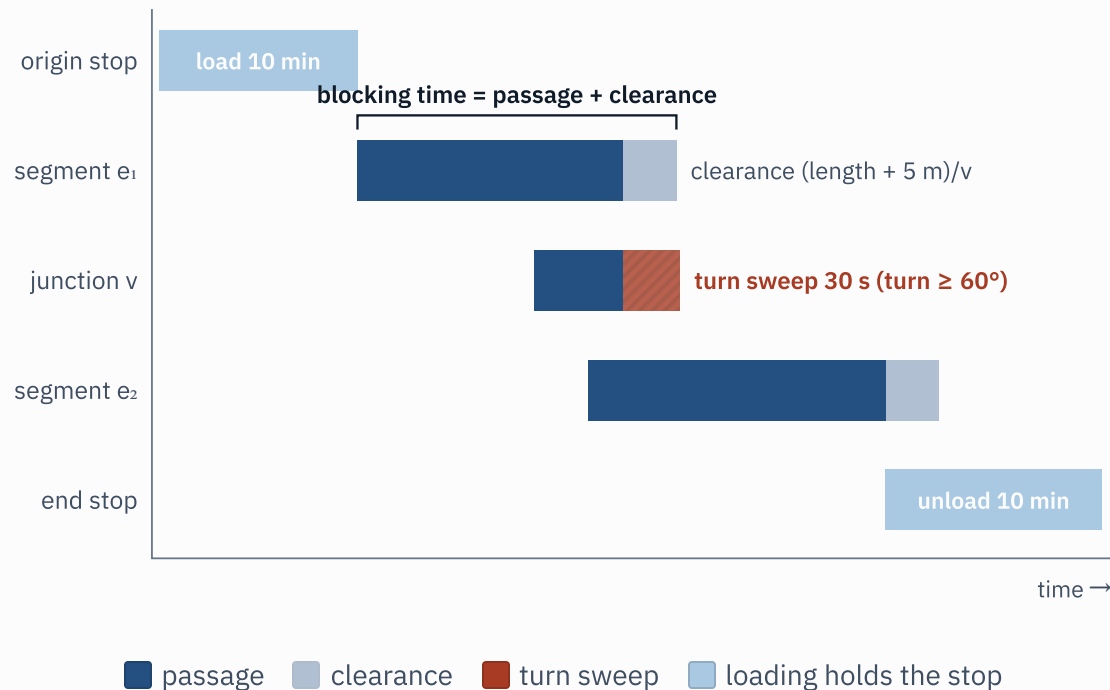
Also feeds

Exchange rate: dual prices are its input; plateau over bound separates it from traffic ordering. Across yards: the map-computed bound and its binding elements are the predictor.

ITEM	STATEMENT (THIS STUDY'S THEOREM LIST)	STATUS
Network bound	Network part of the two ceilings: occupancy-rate LP with width bin-packing rows, crane rows and the window edge term. Holds for rules meeting conditions 1–3; loaded routes fixed	draft; rewriting
Route-free bound	Loaded moves also as multicommodity flow; holds for any routing rule	draft; rewriting
Safe orchestration	Reserving up to the next safe holding point cannot deadlock, with a loss bound relative to the upper bound	being tested
Exchange rate	Given by dual prices; upgrades pay only on tight constraints. Use the route-free bound for new links	in writing
Complexity	Feasibility with deadlines and exclusive resources is NP-hard (reduction from the blocking job shop)	to do
Candidate	Dock-mouth proposition: at nominal load $p < 1$, starvation caused by blocking destabilises the system; waiting off the road or using the dock road in sections restores stability. The other is attainability of the network bound	readings in; to decide

Sources: this study's claims and theorem list (the two-ceilings claim and the theorems quoted in full); status as of 10 Oct 2026 (the route-free bound has a proof draft; the safe-orchestration theorem is tested in the rule comparison; simulation readings are in for the dock proposition); chapter plan of the thesis (8 Oct 2026).

Resources are width containers: segments, junctions and stops are held for a blocking time



- **Three kinds of resources:** segments (721 on Okpo), junctions and entrance nodes (552), drive-in stops (172); with crane cadence, the cranes are dock resources too
- **Capacity** = road width W + lateral clearance c (1 m); a vehicle takes its outline width + c ; at every moment the widths in use stay within capacity
- **Blocking time:** a segment is held from the head entering until the tail clears the next junction, plus longitudinal clearance; a junction for $(D + \text{length} + 5 \text{ m})/v$, plus a 30 s sweep for turns $\geq 60^\circ$
- **Stops:** shops load at their doors (entrance nodes); stockyards and assembly areas have drive-in stops; buffers and docks load at the roadside, so that road segment is the stop

$$\sum_v d_{v,r}(t) \leq C_r \text{ for every resource } r \text{ and time } t$$

Loaded and empty occupancy are asymmetric: passability classes follow from widths, and widths and empty-run rules are scanned as scenarios



■ loaded: block width $B = 10$ m ■ empty: platform $w = 5.5$ m same scale; $c = 1$ m

- **Blocks** drawn with width $B \in \{5, \dots, 15\}$ m and length $\max(B, 10) - 30$ m; loaded outline width $\max(B, 5.5)$ m. Empty platform 16.25×5.5 m; 12 km/h empty, 6 km/h loaded
- **No precise survey:** road widths are graded from imagery (about ± 1.5 m); the three classes follow from widths, not from road-by-road judgement

SCENARIO	OKPO BOUND	YANTAI BOUND
Main scenario	1,720	1,252
Widths +1.5 m	1,789 +4%	1,252 ±0%
Widths -1.5 m	1,674 -3%	1,248 ±0%
Empties exclusive everywhere	1,720 ±0%	1,252 ±0%
Empties pass everywhere	1,822 +6%	1,252 ±0%

Yantai binds at door nodes, so neither widths nor the empty-run rule move it; Okpo binds on narrow roads and responds to both.

Sources: this study's occupancy-model settings and parameter table (block sizes as in the fleet-sizing paper; speeds from the Suzhou Dafang DCY270 data sheet); the table gives the network bound computed in this study, the mean of 10 seeds for Okpo and for Yantai, tasks a day. Sketch drawn to one scale.

Four traffic rules, and the three conditions for the bound

Free flow	No capacity check, vehicles never interact; the field's default. The bound does not apply: one lane exceeds it already at $K = 4$
Whole-route reservation (just-in-time departure)	Empty run, loading, loaded run, unloading booked at once; waits only at stops; deadlock-free by construction
Segment request · reference rule (teleport clearing)	At each junction request “junction + next segment”, else wait on the current segment; a deadlock sends one vehicle off the network for 900 s. Reference only
Safe release in hand	Segment request plus a banker's-algorithm safety check: grants only requests that cannot deadlock, no teleport. 860,291 decisions match a naive reduction; never deadlocked. Conservative: about 2/3 of waits start from a refusal; below reservation for $K \geq 20$

One dispatcher for all four: nearest idle vehicle, first come first served; the fleet bound $K \cdot H / \bar{c}$ holds for all. Best deadlock-free rule in both yards: whole-route reservation; a less conservative one is being tested.

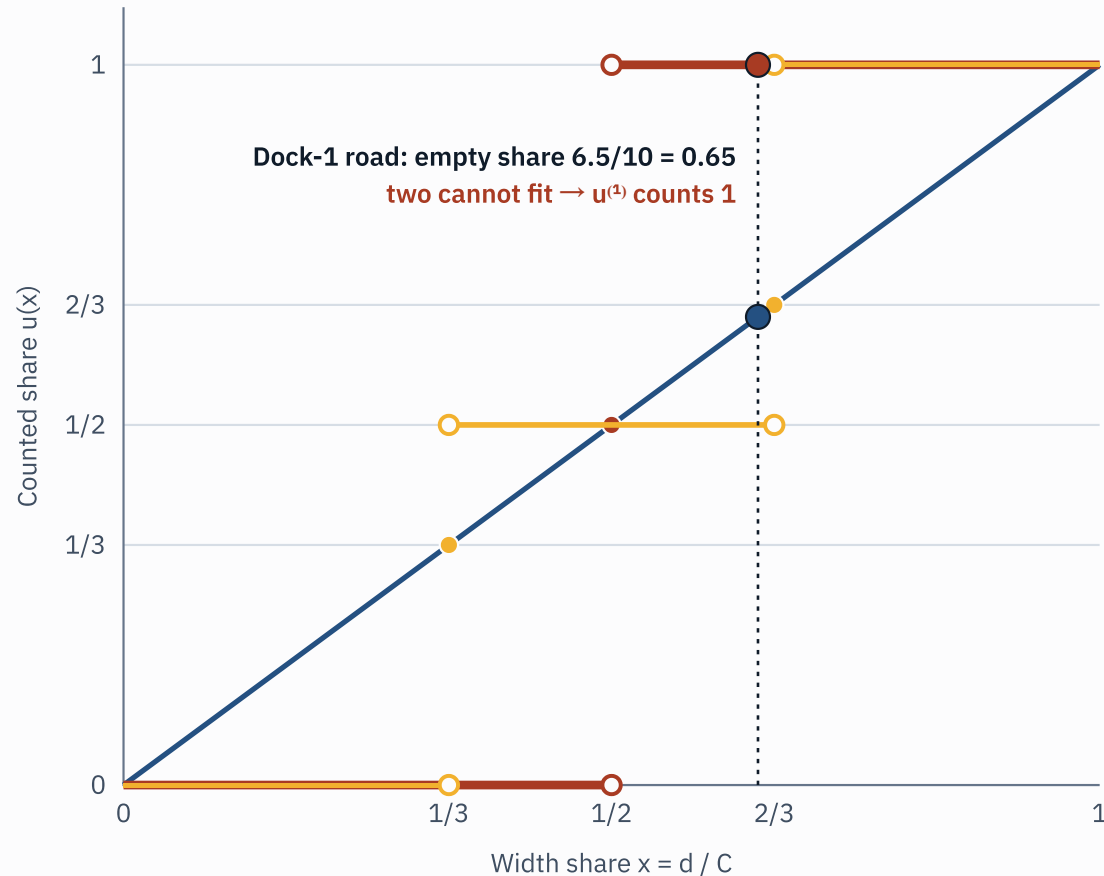
Conditions (proof draft)

- 1** Each task holds each resource for at least its no-wait window, at least at its outline width
- 2** Empty runs occupy resources at least as much as the LP's empty-run coefficients
- 3** The windows counted for one vehicle never overlap: the end window stops when unloading ends
- 1'** Route-free bound: condition 1 on whatever route is taken; the occupancy rules govern requests and releases only, not route choice

Whole-route reservation	proved
Segment request while no clearing occurs	proved
Reference rule: teleports can cut holds short, so condition 1 may fail	checked per case
Safe release: never teleports or deadlocks, i.e. segment request without clearing; waits only lengthen holds	inferred; to confirm
Free flow: no capacity check	does not apply

Sources: the proof draft of the network bound (in preparation; segment request without clearing is inside the proof); this study's occupancy model and the one-lane counterexample; safe release's three-step check, its decision-by-decision comparison with a naive reduction, the small ring counterexample, the share of criterion refusals and the best deadlock-free rule in both yards, all from this study's simulations (Okpo, Yantai); the proof draft of the route-free bound (condition 1').

Network bound: three row types tighten the LP and add the dock



— $u^{(0)} = x$ (basic bound) — $u^{(1)}$: above $1/2 \rightarrow 1$ — $u^{(2)}$: $1/3-2/3 \rightarrow 1/2$

$$\begin{aligned} \max \quad & \lambda \\ \text{s.t.} \quad & \lambda \cdot \bar{a}_r^{(k)} + \sum_a f_a \cdot e_{ar}^{(k)} \leq H \quad \forall r, k = 0, 1, 2 \\ & \sum_{a \rightarrow n} f_a - \sum_{n \rightarrow a} f_a = \lambda \cdot (o_n - d_n) \quad \forall n \\ & \lambda \cdot s_z \cdot 190 \min \leq m_z \cdot H \quad \forall \text{dock } z \end{aligned}$$

Width bin-packing rows

shared widths

The share $x = d/C$ becomes Fekete–Schepers dual-feasible functions $u^{(1)}, u^{(2)}$ (exact rationals): two vehicles that cannot fit side by side each count as exclusive

Crane rows

the dock

Per dock: each erection holds a crane 190 min (hook 10, position and release 180); capacity cranes $\times 16$ h

Window edge term

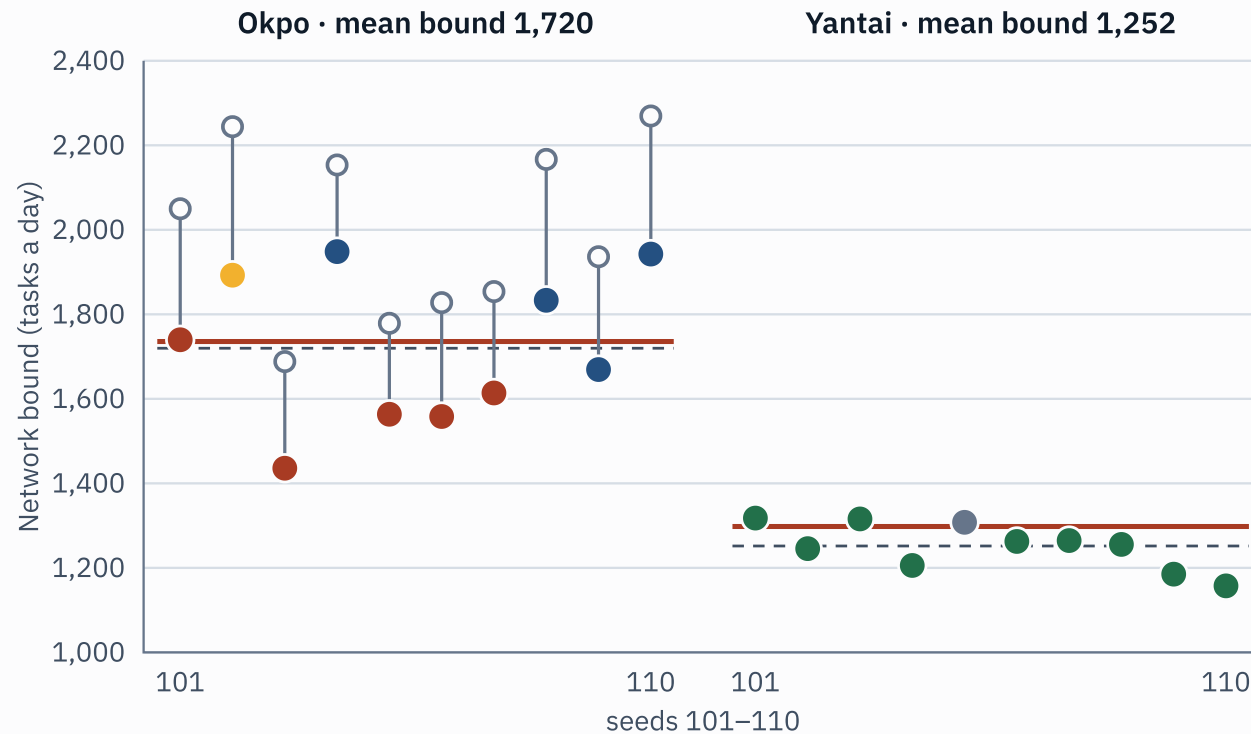
finite window

Over a finite check window W a crane can start once more than its rate: each lift counts $190 \min \times W/(W + 190 \min)$

Theorem: under conditions 1–3, long-run daily throughput $\lambda \leq \min\{\text{network bound}, K \cdot H / \bar{c}\}$; loaded routes on shortest paths, as simulated.

Sources: the proof draft of the network bound (in preparation; lemma 1 is a special case of Fekete & Schepers 2001, proved on its own in the draft); crane rows from this study's crane-cadence settings, the 190 min from Zhang 2025, lines 2525–2528 (another yard); the window edge term serves the case-by-case check over a finite window. $\bar{a}$ and e are the no-wait holds of one task and one empty run on resource r (seconds $\times$ share u); s_z is the erection share to dock z and m_z its cranes. The chart is drawn from the definitions.

Seed by seed in two yards: Okpo tightens by 14% and binds on roads; Yantai binds at the two doors of one building



- basic bound (hollow)
- network bound · dock-1 road
- network bound · through corridor
- network bound · 18 m road
- network bound · paint-shop doors
- network bound · another building's door
- mean of 10 seeds (dashed)
- system bound (solid)

1,997 → 1,720

Okpo main scenario, basic → network bound, mean of 10 seeds (a day, -14%); seeds range 1,436–1,948, system bound 1,736

802 → 703

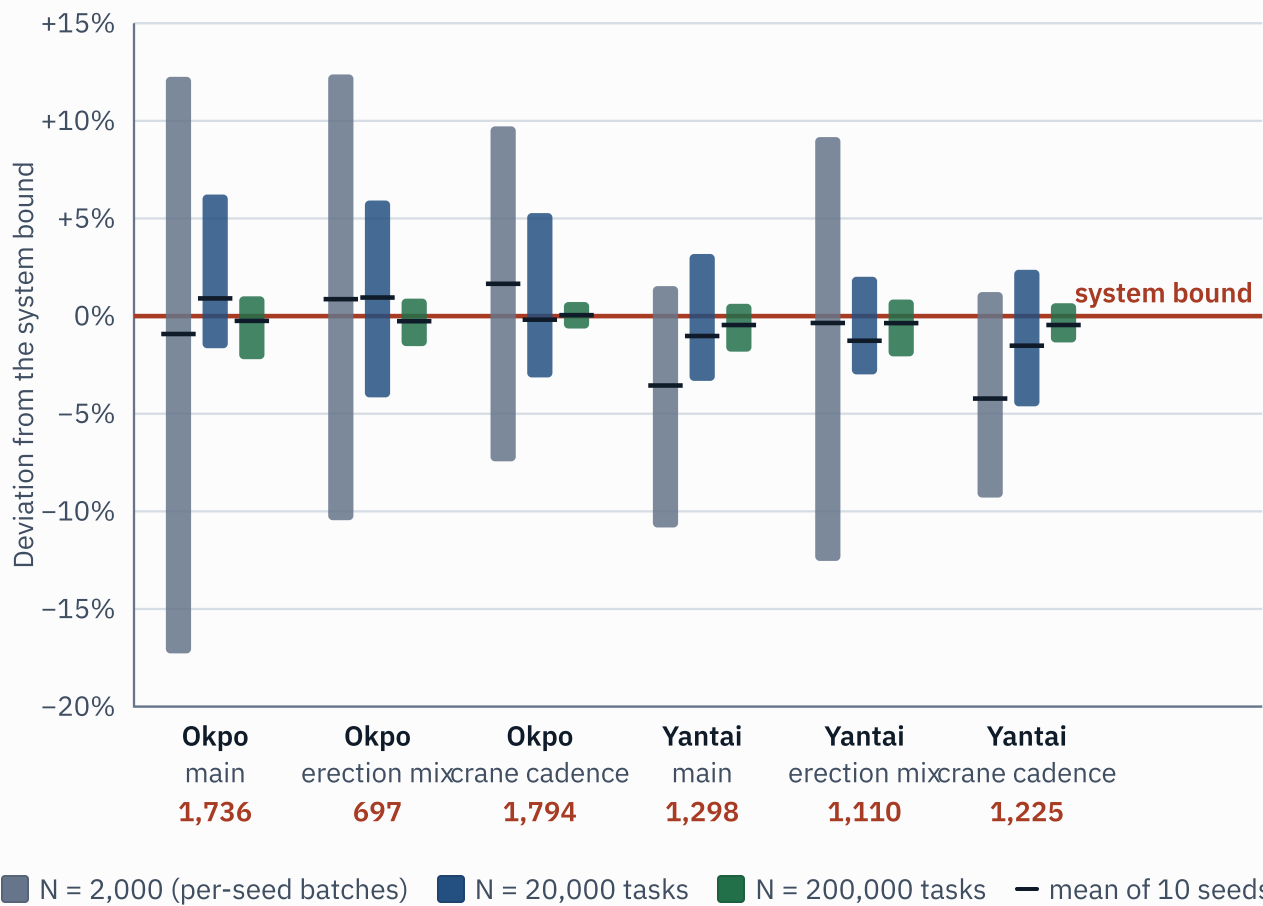
Okpo erection mix (erection at 15%), -12%; 10 of 10 seeds bind on the dock-1 road; system bound 697

1,252

Yantai main scenario, network bound; 9 of 10 seeds bind at the two doors of the paint shop (presumed); system bound 1,298. The basic bound, 1,246, counts the clearance after unloading twice and serves only for comparison

Sources: computations in this study: the Okpo map (before the floating docks and branch roads) and the Yantai map, main scenario and erection mix, 10 seeds each. Conditions: the fleet-sizing paper's flow mix, the first 2,000 tasks of each seed's stream; each road one resource. The system bound (solid line) is computed in this study as well. Yantai's binding nodes share one name in the results and were told apart by rerunning the same code (9 Oct 2026).

The system bound: expected holds from the sampling law take the seed noise out of chapter 3's bound



SCENARIO	OKPO: SYSTEM BOUND · BINDS	YANTAI: SYSTEM BOUND · BINDS
main	1,736	1,298
erection mix	697	1,110
crane cadence	1,794	1,225

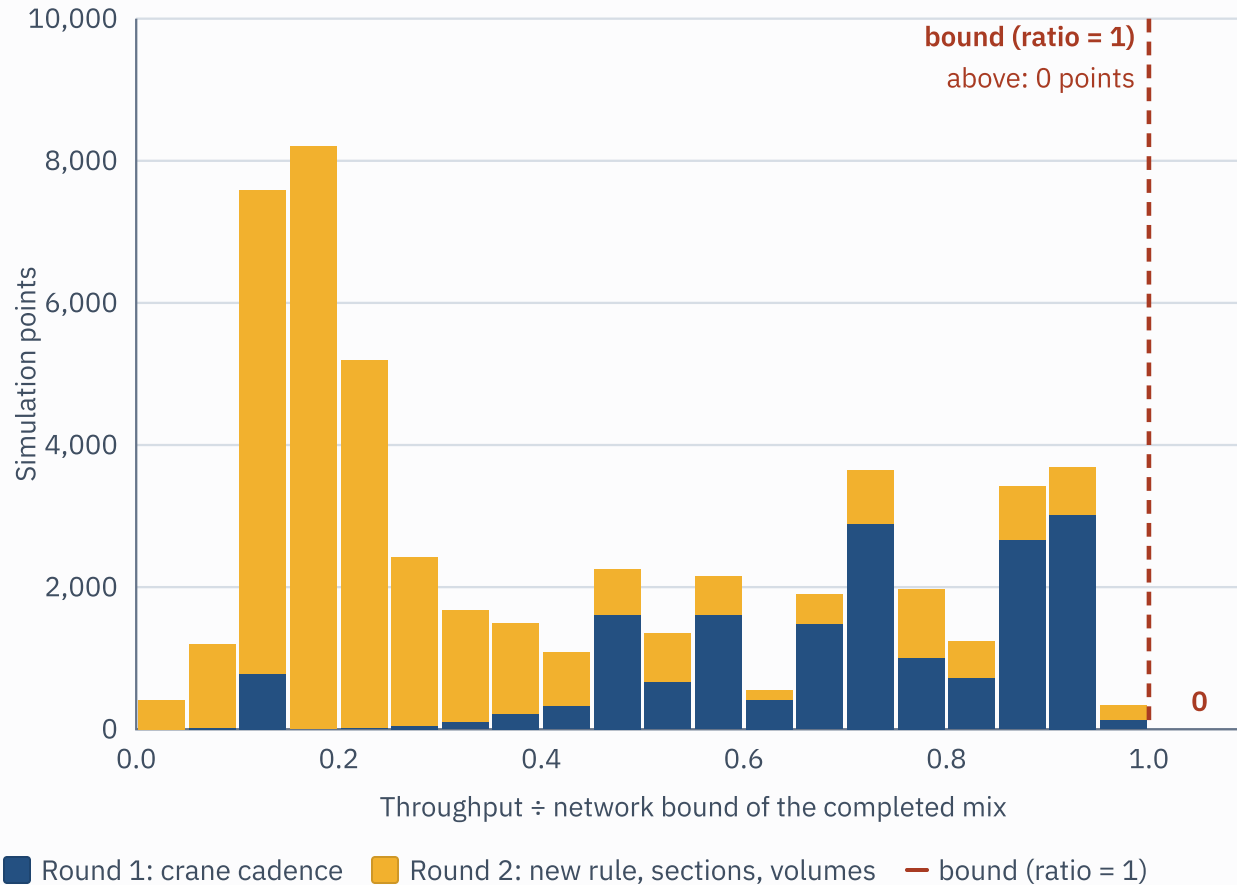
Proposition draft (Jensen)

Without parts fixed in absolute amounts, the reciprocal of the network bound is convex in $(\tilde{a}, \text{on})$ and sampled mixes are unbiased, so the harmonic mean of sampled bounds is, in expectation, at most the system bound. Of 12 readings, 9 lie below the system value and 3 lie above by less than 1.2 standard errors. The formal proof is in preparation.

The same code reproduces the 60 earlier per-seed bounds bit for bit; the system bound solves in under a second. Chapter 3 quotes the system value; per-seed bounds stay in the case-by-case check.

Sources: computations in this study: the system bound and the convergence of sampled bounds at N = 2,000, 20,000 and 200,000 (Okpo and Yantai, 10 seeds each). Conditions: the three scenarios match the per-seed computations item by item (main, erection mix with erection at 15%, crane cadence with erection fixed in absolute amount); each road one resource; the Okpo map before the floating docks and branch roads, the Yantai map. Crane cadence has a part fixed in absolute amount and lies outside the proposition; it is shown for comparison.

Checked case by case: none of 51,767 occupancy simulation points from two rounds exceeds the network bound



0 / 51,767

Above the bound: of 17,760 in round 1 (reservation, reference rule; peak, saturated) and 34,007 in round 2 (plus safe release, sectioned dock road, plant volumes, 16 h / 24 h)

0.989

Largest ratio: round 2, Okpo, $p = 0.99$, sectioned dock road, whole-route reservation, $K = 60$; round 1: 0.973

4,017 / 4,023

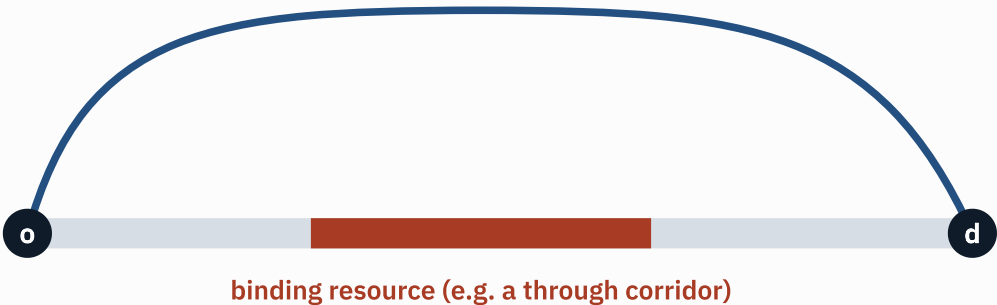
Points with ratio ≥ 0.9 from settings with nominal crane load $p \geq 0.92$: cranes nearly saturated, bound nearly reached

Before: 4 batches when tightening, 12,140 points, largest 0.949; with crane cadence 13 points up to 2.3% over, a window-edge effect, none after the edge term was coded. Round 2 recomputed 410 round-1 points bit for bit (not counted twice).

Sources: simulations in this study, two rounds of case-by-case checks. Round 1: occupancy points of every crane-cadence setting in Okpo and Yantai. Round 2: occupancy points of every batch with the new release rule, the sectioned dock road and plant-level volumes, without free flow and without no-clearing points that complete nothing in the window; sectioned batches use the sectioned network's bound; consistent with the 18,041 + 22,116 points of the round-2 summary (5,740 in both rounds). Ratio = throughput ÷ network bound of the mix completed in that point's window (with crane rows and the edge term).

The route-free bound: loaded moves as multicommodity flow, valid for any routing rule

Route-free bound: the LP picks routes (column generation on duals)



Network bound: loaded moves take the simulation's shortest path

max λ

s.t. $\lambda \cdot F_{\mathcal{I}} + \sum_P y_P \cdot c_{P,\mathcal{I}} + \sum_a f_a \cdot \tilde{e}_{a\mathcal{I}} \leq H \quad \forall \mathcal{I}, k = 0, 1, 2$

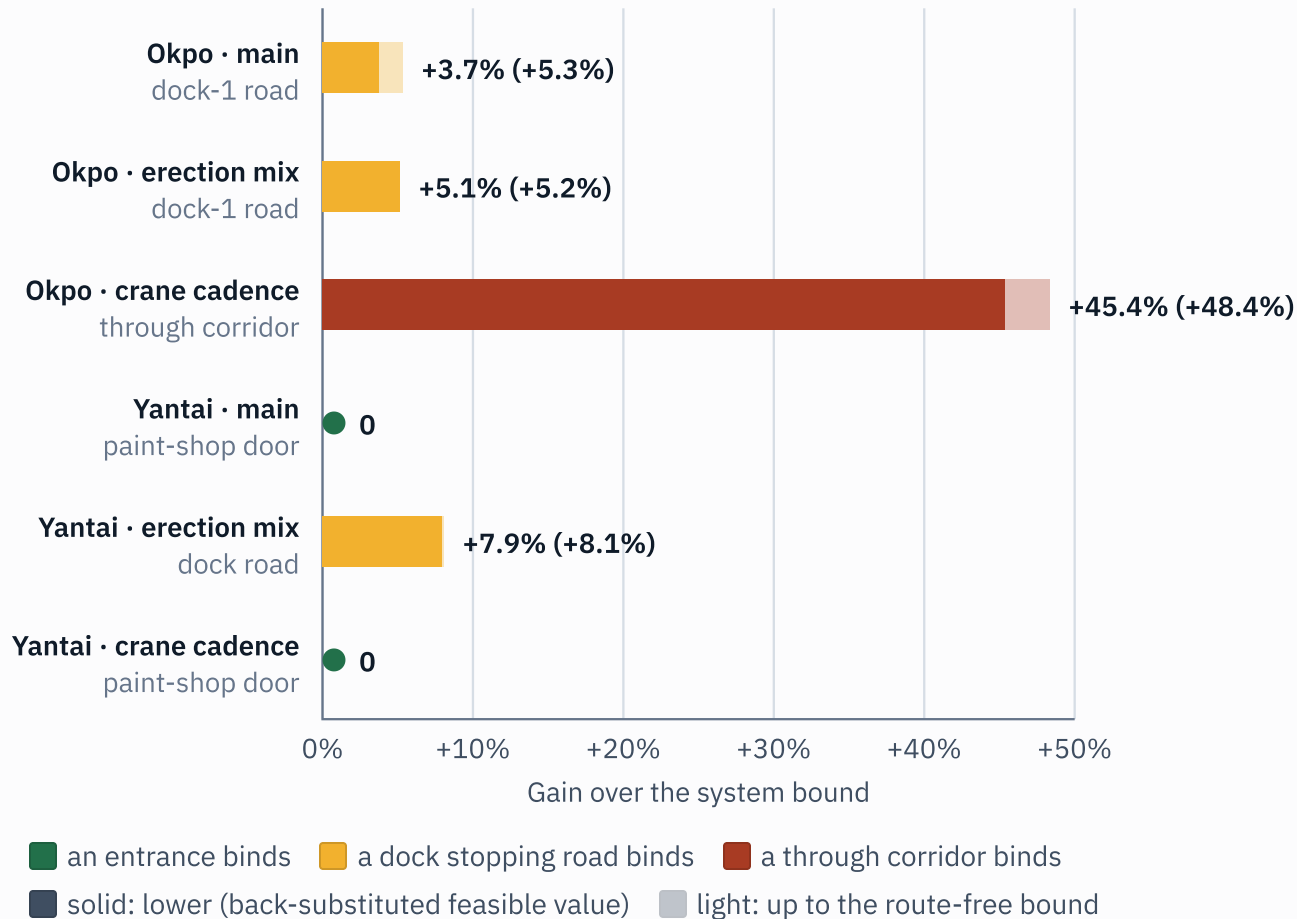
$\sum_{P \in \mathcal{P}(o,d)} y_P = \lambda \cdot q_{odb} \quad \forall (o, d, b)$

empty-run flow conserved, as in the network bound

Counted exactly route-independent	Loading and tail clearance at the origin, unloading, crane rows, width shares, empty runs (as in the network bound)
Lower bounds route-dependent (seven relaxations)	Vehicle length at the platform's 16.25 m, no turn sweep, the end stop held only from the last arc, tails cut at the return distance
Certificate least bound over rounds	μ truncated at zero, v repaired to strict dual feasibility by one super-source Dijkstra: independent of solver tolerance and of convergence
Checks cross-check	network bound $\leq R \leq$ route-free bound on 60 seeds; no counterexample: a direct arc-flow solve differs by 4.6e-15, 80 random duals, 32,191 detour columns checked resource by resource
Route-free theorem (suggested wording): for traffic rules meeting conditions 1', 2 and 3 and any routing rule, $\lambda \leq \min\{\text{route-free bound}, K \cdot H / \bar{c}\}$. The draft proof is being rewritten. The largest LP has 23,316 rows and 29,917 columns; column generation takes 8 rounds, about 6 min.	

Sources: the proof draft of the route-free bound (in preparation; the route-free LP, the seven relaxations, condition 1', the Lagrangian certificate); computations in this study (Okpo, Yantai, 60 seeds) and a cross-check (direct arc-flow solve, random duals, detour columns checked resource by resource). R: the value with routes fixed to shortest paths and the same relaxations as the route-free bound. Sketch not to scale.

Route slack in the bound as an interval: zero at an entrance, +4–8% on a dock stopping road, +45% on a through corridor



Entrance

Yantai main, crane

Zero by construction: the binding entrance has a single incoming arc, so its holds come from handling and the one last stretch, the same on any route

Dock stopping road

Okpo dock 1, Yantai dock

Small through share (0.3–0.7%; 20%). The slack comes mainly from local flow entering the stops from the other end, plus a little through traffic going round

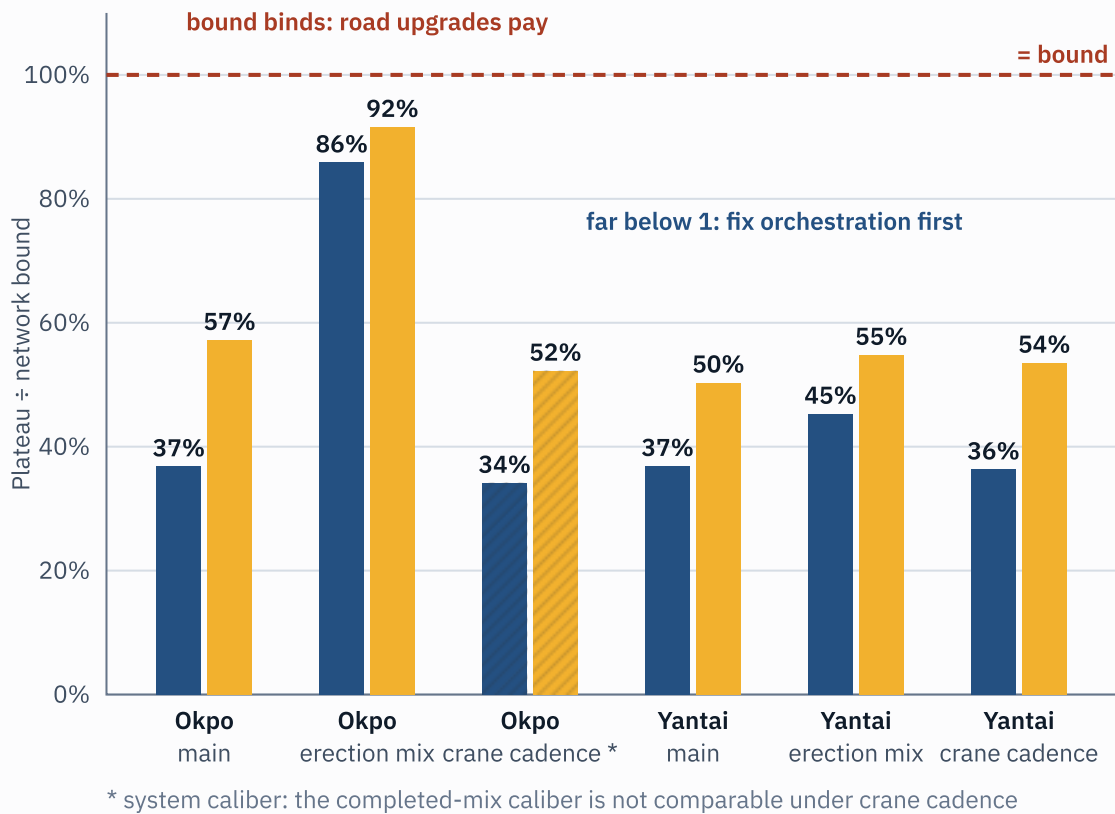
Through corridor

Okpo, crane cadence

97% of its 462 loaded moves a day pass through; the detour around it still has 22% to spare and adds 235 m on average

Lower = back-substituted feasible value – system bound: keep the route-free route split, restore every coefficient to the unrelaxed form and re-optimize empty runs; upper = route-free bound – system bound. It caps what detour orchestration could recover, and chapter 4 measures how much it does; each class has only 1–3 points, and the retrofit simulations add more before the wording becomes a mechanism.

Two uses of the bound: plateau over bound says reorder or upgrade; two dual-price columns give the exchange rate



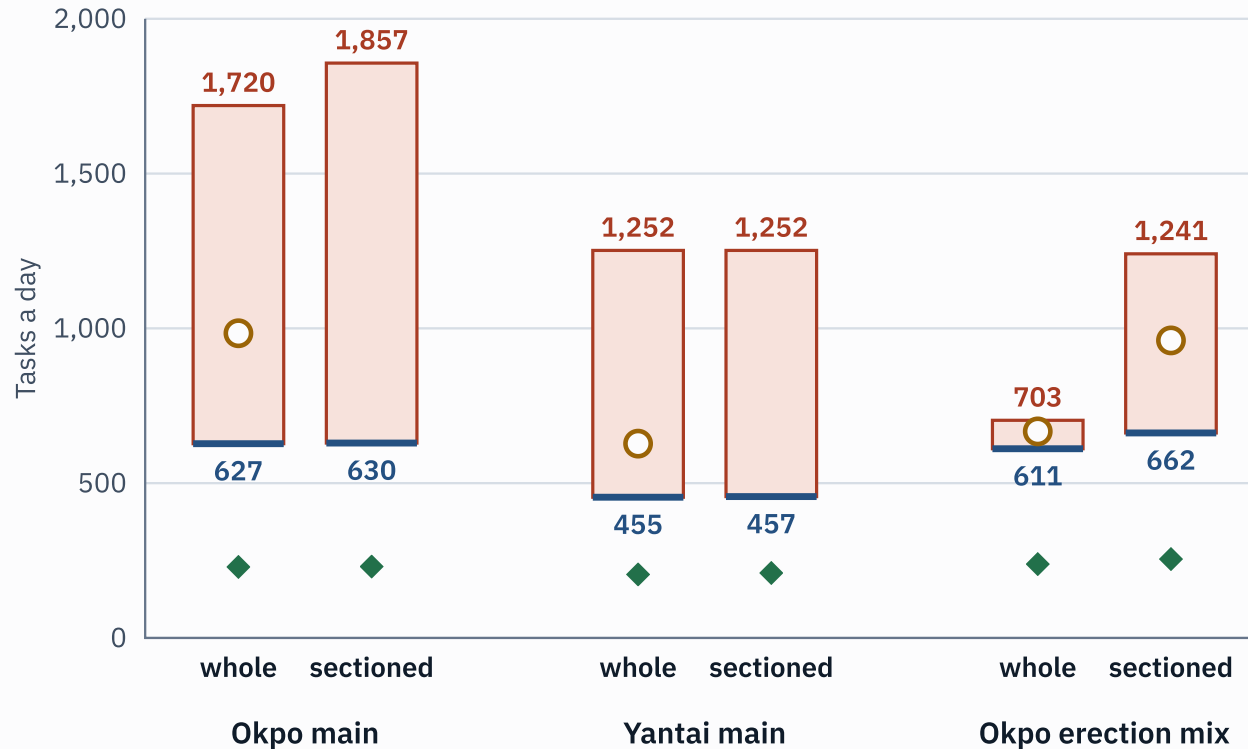
■ whole-route reservation ■ reference rule (teleport clearing)

Sources: simulations and computations in this study (Okpo, Yantai). Plateau = mean saturated throughput over K = 100–150 (shortest-path routing); divided by the network bound of the batch's completed mix, or by the system bound for Okpo's crane cadence (the completed-mix caliber scales erection up and reads as crane utilisation). The two doors of Yantai's paint shop (presumed) are both full: upgrading one barely helps, both together give +144 at first order.

RESOURCE · SCENARIO	SHORTEST PATHS (NETWORK)	DETOURS (ROUTE-FREE)
dock-1 road Okpo · main	holds –10%: +134 (first-order +193) widen 1 m: 0	holds –10%: +203 widen 1 m: 0
through corridor Okpo · crane cadence	holds –10%: +198 widen 1 m: +188	0
detour corridor Okpo · crane cadence	0	holds –10%: +283 (first-order +295) widen 1 m: +164
paint-shop door Yantai · main	holds –10%: +2.3 (first-order +144)	same (entrances do not depend on routes)
dock road Yantai · erection mix	holds –10%: +123 widen 1 m: +127	holds –10%: +133 widen 1 m: +138

Tasks a day. “Holds –10%” = 1.78 extra hours a day; re-solved values, first-order estimate in brackets where it differs. Where the two columns disagree, road upgrades and orchestration can stand in for each other; width pays in steps, so the retrofits price “widen to the next class”.

Sectioning the dock road: in the main scenario only the upper edge rises (+8%); where the dock binds both edges move (+76.5% / +8.3%)



■ interval: reservation plateau → bound ◆ safe-release plateau

○ reference rule (teleport clearing) plateau

- **Okpo main:** bound 1,720 → 1,857 (+8%), binding shifts from two roads in turn to one corridor; reservation plateau 627 → 630 (+0.3%): the rule, not this road, limits throughput
- **Okpo erection mix:** bound 703 → 1,241 (+76.5%), reservation 611 → 662 (+8.3%); plateau over bound 0.87 → 0.53: change the granularity and the bottleneck moves from the road to the rule
- **Yantai:** an entrance binds; bound 1,252 either way
- **Still a bound after sectioning:** resources are only finer, conditions 1–3 still hold, the proof is unchanged; on one mix the sectioned bound is at least the whole-road one

The lower edge is the best deadlock-free rule tested so far: whole-route reservation. Safe release plateaus at 205–255, peaking at $K \approx 18$ –31, below reservation throughput. The 12 m widening (1,805 / 786) and passing bays of an earlier probe get ten-seed numbers in the retrofits.

Sources: simulations in this study: Okpo and Yantai at both granularities, with whole-road reservation and the reference rule reused from earlier batches. Conditions: the fleet-sizing paper's flow mix, mean over saturated $K = 100$ –150, 10 seeds, 4 days; sectioning = block boundaries 25 m either side of every stop on roads longer than 200 m with roadside stops (exactly the 4 dock roads of the two yards), a scenario setting; the reference rule has direction locks when sectioned; the Okpo map before the floating docks and branch roads, the Yantai map. The widening probe is an earlier exploration (9 Oct 2026).

Both yards' binding elements sat on conservative modelling choices: the key-element review becomes a fixed step

Okpo · the dock-1 road (9 m wide, 1,095 m)

Main scenario: binds in 5 of 10 seeds, dual price 98.9 a day per extra hour (about 4.2 vehicles); the through corridor in 4, 115.5. Erection mix: 10 of 10, 43.9 (1.8 vehicles).

Imagery check: with a vehicle unloading or waiting for the crane at the dock edge, the neighbouring assembly areas stay reachable. → Okpo's main scenario is now sectioned at the stops (bound 1,857).

Yantai · the two doors of the paint shop (presumed)

Binds in 9 of 10 seeds. Both doors are dead ends where vehicles load and unload, holding the door about 10 min; with only two doors, each serves about 7% of all tasks.

Imagery check: two doors with room to pass in front, so exclusive loading at the door is conservative. Price table: both doors full (1.000 / 0.998); one door adds 2.3 a day, both +144 at first order. → Retrofits: a door-granularity scenario, the two doors as one upgrade.

The fixed step (every yard in chapter 6)

- 1 Compute the system bound and every resource's dual price (shortest-path dispatch and detours allowed)
- 2 List the three elements with the highest dual prices
- 3 Check on imagery: **road width, number of entrances, how traffic passes a stop**
- 4 Choose the resource granularity, recompute, and record it in the map's source field

This is a pinpoint check of a few key elements, not a precise survey; it can go into the thesis as an engineering method (minimal human check).

What stands and what comes next: every computation lands in a figure, a table or a section of the thesis

In hand

- ✓ The network-bound LP with its three kinds of rows, code, tests and a proof draft
- ✓ The system bound and the route-free bound with its certificate; route slack as an interval
- ✓ 51,767 occupancy points checked, none over (two rounds)
- ✓ Safe release exact and never deadlocked; proof unchanged with the dock road sectioned
- ✓ Citations checked; the nearest same-field work told apart in full text

Conditions (stated once): the system and route-free bounds bound the steady state; the case check uses each point's completed mix; the network bound fixes loaded routes to shortest paths, the route-free bound frees them; the reference rule is outside the proof; both bounds treat each road whole, Okpo's main scenario is sectioned (1,857); 16-h day; the fleet-sizing paper's flow mix; the Okpo map before floating docks and branch roads, the Yantai map; later work uses the Okpo map with them.

WORK	WHAT	OUTPUT (PLACE IN THE THESIS)	STATUS
Rules	A less conservative deadlock-free rule: the banker's check plus step-wise feasibility search, or claims cut at the next safe holding point (safe orchestration); re-checks the lower edge on the new Okpo map	Rule comparison (ch. 4); lower edge of the interval table (ch. 3)	running
Retrofits	Upgrade simulations and the first exchange-rate table; door granularity; capacity against block length; the three-resource dock network	Rate table (ch. 5); Yantai's bound and binding elements; the dock section	next
Floating docks	Floating docks; a finite holding area S	Vehicle-type case; holding areas (ch. 5)	after that
Proofs	Rewrite the proofs of both bounds; settle the wording of the Jensen proposition and the three readings	Chapter 3, theorems and method	weeks 3–4
–	Candidate: attainability of the network bound (a periodic rule reaching $1 - \epsilon$ of it when the dock dominates)	Chapter 3, theorems	to do
–	Complexity: NP-hard (from the blocking job shop)	Chapter 3, theorems	to do

Sources: this study's work plan (decided on 10 Oct 2026: the rule comparison first, then the retrofit simulations); this study's claims and theorem list. Running and queued work are plans, not results.

Risks and mitigations: each has a way to strengthen it and a place in the thesis; the plan stands

RISK	MITIGATION (HOW TO STRENGTHEN IT)	OUTPUT
The route-free bound takes the vehicle length at its lower bound; on detours this matters about as much as R minus the network bound	Back-substitution already gives the lower value; split commodities into 3 length classes (mean shortfall 5.2 m → 1.5 m), warm-started from the unsplit column pool	Narrower route-slack interval (ch. 3); an appendix table
Degenerate optima put the price on a single resource	Re-solves already give exact gains; the retrofit simulations re-solve combined upgrades (both paint-shop doors) and run a relax-and-find-the-next-bottleneck sequence	Combined-upgrade rows of the rate table (ch. 5)
Safe release is conservative (about 2/3 of waits start from refusals; a small ring counterexample), so the lower edge is reservation for now	Rule comparison: step-wise feasibility search or claims cut at the next safe holding point, reporting the refusal share and feeding the same batches into the interval table	Rule comparison (ch. 4); the lower edge
The 200 m and 25 m of sectioning are scenario settings	Check Okpo's dock-1 road and Yantai's two dock roads on imagery; a block-length scan in the retrofit simulations turns the 25 m into a curve	Scope line of the dock section; “control granularity” (ch. 5)
Only 1–3 points per binding-resource class	Every upgrade scenario reports the through share, the alternative route's slack and the route-slack interval, drawn as a scatter	“Upgrade the road or the orchestration” (ch. 5)

Sources: this study's risk analysis and strengthening plans for the system bound, the route-free bound and safe release. None changes the plan.

What is new: route-fixed and route-free LP bounds for networks with shareable widths and asymmetric occupancy

- **Within the search** (literature search of 9 Oct 2026), the shipyard transport literature has no capacity bound; the nearest same-field work, checked in full text (right), does not narrow the claim, and the introduction names it
- This study extends the occupancy-rate LP to road networks with **shareable widths and asymmetric loaded and empty occupancy**, valid for any rule meeting the three conditions
- Dual-feasible functions serve as **time-integrated capacity rows**; crane rows and the window edge term bring the dock into the same bound
- **A route-fixed bound beside a route-free one**, separating the route slack and its sources; whether the field already has a route-free bound under occupancy is still to be searched
- **As a system**: tracing on public imagery → bound → dual prices (two columns) → key-element review, one LP per yard

Nearest same-field work: Tao et al. 2017 (full text)

ITEM	TAO ET AL. 2017	THIS STUDY
Road capacity	in vehicles: 2 on main roads, 1 on side roads	width containers and blocking times that change with load
Traffic rules	one: first come first through, wait or reroute	four rules side by side, incl. deadlock-free release
Deadlock	not handled	reservation and safe release, deadlock-free by construction or by test
Capacity bound	none, simulation only	network, system and route-free bounds, checked case by case
Yards	one, tuning vehicle types by load	two, traced on public imagery

Sources built on: same-mechanism work from other fields supports and informs this study; it is not prior coverage

SOURCE BUILT ON	WHAT THIS STUDY TAKES	READ
Railway blocking times, UIC 406; Goverde 2007	Blocking time $\rightarrow$ occupancy rows; max-plus gives the cycle of a given order, a lower-side result	abstract; UIC 406 to check
Jensen, Schmidt, Nielsen 2020	Railway network capacity without a fixed timetable, computed by branch and bound: structurally among the closest; “ignoring junction conflicts overstates capacity” matches junctions as exclusive resources	full text
Campos, Chiola, Silva 1991	LP throughput bounds for Petri nets whose conflict decisions keep relative frequencies, the same structure as the network bound's fixed routes; the route-free bound frees the routes and goes beyond that class	full text
Queueing networks with blocking: Onvural 1990; Balsamo et al. 2001	Theory of blocking and starvation	abstract; book to check
Fekete & Schepers 2001	Dual-feasible functions (bin-packing lower bounds) $\rightarrow$ width bin-packing rows	lemma proved in the draft
The banker's algorithm (Dijkstra); Habermann 1969	The safe-state test $\rightarrow$ safe release	to check
Mascis & Pacciarelli 2002	Blocking job shop $\rightarrow$ complexity reduction	abstract

Sources: literature search and reading notes (9 Oct 2026; Campos et al. and Jensen et al. read in full); the original source of the banker's algorithm is still to be checked; “max-plus is on the lower side” is argued in the proof draft of the network bound. The phrasing is “we extend X to ...” and “building on X ...”.

Target journals for paper 1 (chapters 2–3): class B first, class A as a stretch

TIER	JOURNAL (SCHOOL CLASS)	WHY IT FITS / CONDITION
First	Transportmetrica A: Transport Science (class B)	Transport science at the core: the occupancy model, a rule-free bound and the capacity interval, with rigorous modelling and strong baselines
First	Ocean Engineering (class B)	Shipyards transport mechanisms and resource limits as the main line: two yards, tracing on public imagery, the dock and erection
Stretch	Reliability Engineering & System Safety (class A)	Condition: the dock-mouth proposition is written and proved as a stability or resilience result, “blocking-induced starvation, instability and recovery”
Alternative	IEEE Transactions on Automation Science and Engineering (class B)	Capacity and release rules of automated logistics systems (safe release, reservation)
Alternative	Applied Mathematical Modelling (class B)	The model and bounds themselves: width bin-packing rows, the system bound, the route-free bound and its certificate
Supplementary	Computers & Industrial Engineering; Journal of Ship Production and Design (class C)	Well-matched supplementary outputs

Class B first; class A as a stretch, with its condition in the table. Graduation needs at least one paper in a class-B-or-above journal (the segmentation paper is already published in Ocean Engineering, class B).

Sources: the School of Shipbuilding Engineering 2026 journal directory (graduation requirements file, version of 23 Sep 2026), §1, §2, §7; the paper plan of 10 Oct 2026 (paper 1 covers the network-bound, capacity-interval and dock sub-studies). “Why it fits” is this study’s own matching of the sub-study to each journal, to be checked against each journal’s aims and scope before submission.

References and data sources

1. Balsamo, S., de Nitto Personè, V., Onvural, R. (2001). *Analysis of Queueing Networks with Blocking*. Kluwer, Boston. (to be checked)
2. Campos, J., Chiola, G., Silva, M. (1991). Ergodicity and throughput bounds of Petri nets with unique consistent firing count vector. *IEEE Transactions on Software Engineering* 17(2), 117–125. doi:10.1109/32.67593
3. Fekete, S.P., Schepers, J. (2001). New classes of fast lower bounds for bin packing problems. *Mathematical Programming* 91(1), 11–31. doi:10.1007/s101070100243
4. Goverde, R.M.P. (2007). Railway timetable stability analysis using max-plus system theory. *Transportation Research Part B* 41(2), 179–201. doi:10.1016/j.trb.2006.02.003
5. Habermann, A.N. (1969). Prevention of system deadlocks. *Communications of the ACM* 12(7), 373–377. doi:10.1145/363156.363160
6. Jensen, L.W., Schmidt, M., Nielsen, O.A. (2020). Determination of infrastructure capacity in railway networks without the need for a fixed timetable. *Transportation Research Part C* 119, 102751. doi:10.1016/j.trc.2020.102751
7. Mascis, A., Pacciarelli, D. (2002). Job-shop scheduling with blocking and no-wait constraints. *European J. Operational Research* 143(3), 498–517. doi:10.1016/S0377-2217(01)00338-1
8. Onvural, R.O. (1990). Survey of closed queueing networks with blocking. *ACM Computing Surveys* 22(2), 83–121. doi:10.1145/78919.78920
9. Reveliotis, S.A. (2000). Conflict resolution in AGV systems. *IIE Transactions* 32(7), 647–659. doi:10.1080/07408170008967423
10. Tao, N. 陶宁蓉, Jiang, Z. 蒋祖华, Liu, J. 刘建峰, Yang, Z. 杨震 (2017). 考虑多平板车的船厂分段运输问题仿真建模与分析. *河北工业科技* 34(6), 389–394. doi:10.7535/hbgykj.2017yx06001
11. UIC (2013). *UIC Code 406: Capacity*, 2nd ed. International Union of Railways, Paris.
12. Zhang, S. 张水根 (2025). 大型船海装备制造场地智慧调度与优化系统设计及实现. 硕士学位论文, 沈阳工业大学. doi:10.27322/d.cnki.gsgyu.2025.000182

Data and documents: simulation and computation results of this study (Okpo and Yantai, with conditions in each slide's source line); proof drafts of the network and route-free bounds (in preparation); this study's claims and theorem list; literature search and full-text checks (9 Oct 2026); the school's 2026 journal directory; the DCY270 data sheet. Chart numbers are generated directly from this study's results.

What this sub-study gives the thesis: a network bound that can be computed, checked and that points the way

1

Computable

One LP per yard: the system bound in under a second, the route-free bound by column generation in at most about 6 min; only networks traced on public imagery and public parameters.

2

Checked

Proof drafts for both bounds, with a certificate for the route-free one; none of 51,767 occupancy simulation points exceeds the bound; a cross-check found no counterexample to the route-free bound.

3

Points the way

Dual prices in two columns name the binding roads and doors, and show where road upgrades and orchestration can stand in for each other.

[← Back to the thesis briefing](#)[Capacity interval →](#)[Dock and erection peak →](#)[Fleet-network co-design \(exchange rate\) →](#)

Transporters own the road: capacity is capped by the network, and the bound shows how high the cap is and where it binds.

Thank you.